QUESTION:1

```
#include <iostream>
using namespace std;
class student{
    public:
    string name;
    int roll_no;
    float marks;
    void display_info(student s);
};
void student::display_info(student data){
    cout<<"=====DISPLAY STUDENT DATA======"<<endl;
    cout<<"NAME: "<<data.name<<endl;
    cout<<"ROLL NUM: "<<data.roll_no<<endl;
    cout<<"MARKS: "<<data.marks<<endl;
}
int main(){
   student data;
   cout<<"NAME: ";
   getline(cin,data.name);
   cout<<"ROLL NUM: ";
   cin>>data.roll_no;
   cout<<"MARKS: ";
   cin>>data.marks;
   data.display_info(data);
  return 0;
}
```

```
NAME: Waleeja
ROLL NUM: 05
MARKS: 23
=====DISPLAY STUDENT DATA======
NAME: Waleeja
ROLL NUM: 5
MARKS: 23
```

QUESTION:2

```cpp
#include <iostream>
using namespace std;
class book{
    public:
    string book_title;
    string author_name;
    int price;
    void display_details(book b1);
};

void book::display_details(book book1){
     cout<<"NAME: "<<book1.book_title<<endl;
     cout<<"ROLL NUM: "<<book1.author_name<<endl;
     cout<<"MARKS: "<<book1.price<<endl;
}

int main(){
    book b1,b2,b3;
    //book 1
    cout<<"BOOK 1 TITLE: ";
    getline(cin,b1.book_title);
    cout<<"AUTHOR NAME: ";
    cin>>b1.author_name;
    cout<<"PRICE: ";
    cin>>b1.price;
```

```cpp
cout<<"BOOK 2 TITLE: ";
cin.ignore();
getline(cin,b2.book_title);
cout<<"AUTHOR NAME: ";
cin>>b2.author_name;
cout<<"PRICE: ";
cin>>b2.price;


//book 3
cout<<"BOOK 3 TITLE: ";
cin.ignore();
getline(cin,b3.book_title);
cout<<"AUTHOR NAME: ";
cin>>b3.author_name;
cout<<"PRICE: ";
cin>>b3.price;
cout<<"===BOOK 1 DETAILS==="<<endl;
b1.display_details(b1);
cout<<"===BOOK 2 DETAILS==="<<endl;
b2.display_details(b2);
cout<<"===BOOK 3 DETAILS==="<<endl;
b3.display_details(b3);
return 0;
```

```
BOOK 1 TITLE: pride and prejudice
AUTHOR NAME: jane
PRICE: 1500
BOOK 2 TITLE: alice in wonderland
AUTHOR NAME: lewis
PRICE: 1200
BOOK 3 TITLE: jane eyre
AUTHOR NAME: charlotte
PRICE: 1350
===BOOK 1 DETAILS===
NAME: pride and prejudice
ROLL NUM: jane
MARKS: 1500
===BOOK 2 DETAILS===
NAME: alice in wonderland
ROLL NUM: lewis
MARKS: 1200
===BOOK 3 DETAILS===
NAME: jane eyre
ROLL NUM: charlotte
MARKS: 1350
```

QUESTION:3

```cpp
#include <iostream>
using namespace std;
class rectangle{
  public:
    int length;
    int width;
    void calculateArea(rectangle val);
};
void rectangle::calculateArea(rectangle val){
     int l=val.length;
     int w=val.width;
     int rect;
     rect=l*w;

     cout<<"AREA = "<<rect;
}

int main(){
    rectangle data;
    cout<<"LENGTH: ";
    cin>>data.length;
    cout<<"WIDTH: ";
    cin>>data.width;
    data.calculateArea(data);
}
```

```
LENGTH: 3
WIDTH: 4
AREA = 12
c:\Sem 2\OOP LABS\lab6>
```

QUESTION:4

```cpp
#include <iostream>
using namespace std;
class result{

    string name;
    string roll_no;
    int marks[5];
public:
    void inputData(){
        cout<<"NAME: ";
    cin>>name;
    cout<<"ROLL_NUM: ";
    cin>>roll_no;
    cout<<"MARKS: ";
    for(int i=0;i<5;i++){
       cin>>marks[i];
       if(marks[i]<0 || marks[i]>100){

        i--;
       }
    }
    }
```

```
float calculateAverage() {
    float avg;
    int sum = 0;
    for (int i=0;i<5;i++){
        sum += marks[i];
    }

    avg=sum / 5.0;
    return avg;
}

    void displayResult() {
    cout << "====RESULT===="<<endl;
    cout<<"Name: "<<name<<endl;
    cout<<"ROLL NUM: "<<roll_no<<endl;
    cout<<"MARKS: "<<endl;
    for(int i=0;i<5;i++) {
    cout<<marks[i]<<" ";
    }
    cout<<endl;
    float avg = calculateAverage();
    cout<<"AVERAGE:"<<avg<<endl;
    cout<<"STATUS: ";
```

```
        cout<<"AVERAGE:"<<avg<<endl;
        cout<<"STATUS: ";
        if(avg>=50)
        cout<<"PASS"<<endl;
        else
        cout<<"FAIL"<<endl;
    }
};

int main(){
    result std1, std2;
    cout<<"Enter details for Student 1: "<<endl;
    std1.inputData();
    cout<<"Enter details for Student 2: "<<endl;
    std2.inputData();

    std1.displayResult();
    std2.displayResult();

    return 0;
}
```

```
Enter details for Student 1:
NAME: waleeja
ROLL_NUM: 23
MARKS: 45
56
67
78
89
Enter details for Student 2:
NAME: ali
ROLL_NUM: 34
MARKS: 54
43
32
21
65
```

```
====RESULT====
Name: waleeja
ROLL NUM: 23
MARKS:
45 56 67 78 89
AVERAGE:67
STATUS: PASS
====RESULT====
Name: ali
ROLL NUM: 34
MARKS:
54 43 32 21 65
AVERAGE:43
STATUS: FAIL
```

QUESTION:5

class_header.h | function.cpp | main.cpp

lab6 > main.cpp > ...

```cpp

/*QUESTION 5*/
#include <iostream>
#include "class_header.h"
using namespace std;
int main(){
    employee emp1, emp2;

    cout << "Enter details for Employee 1:\n";
    emp1.inputData();
    cout << "Enter details for Employee 2:\n";
    emp2.inputData();

    emp1.displayDetails();
    emp2.displayDetails();

    return 0;

}
```

class_header.h | function.cpp | main.cpp

lab6 > function.cpp > ...

```cpp
#include <iostream>
#include "class_header.h"
using namespace std;
void employee::inputData(){
    cout<<"Enter Name: ";
    cin>>name;
    cout<<"Enter Employee ID: ";
    cin>>employee_id;
    cout<<"Enter Basic Salary: ";
    cin>>basic_salary;
}
int employee::calculateBonus() {
    int bonus=basic_salary * 0.10;
    return bonus;
}
void employee::displayDetails() {
    int bonus = calculateBonus();
    double finalSalary = basic_salary + bonus;
    cout<<"===EMPLOYEE DETAILS==="<<endl;
    cout<<"NAME: "<<name<<endl;
    cout<<"EMPLOYEE ID: "<<employee_id<<endl;
    cout<<"BASIC SALARY: "<<basic_salary<<endl;
    cout<<"BONUS: "<<bonus<<endl;
    cout<<"FINAL SALARY: "<<finalSalary<<endl;
}
```

class_header.h | function.cpp | main.cpp

lab6 > class_header.h > ...

```cpp
#include <iostream>
using namespace std;
class employee{
    string name;
    int employee_id;
    int basic_salary;
    public:
    void inputData();
    int calculateBonus();
    void displayDetails();

};
```

```
Enter details for Employee 1:
Enter Name: waleeja
Enter Employee ID: 35
Enter Basic Salary: 40000
Enter details for Employee 2:
Enter Name: areej
Enter Employee ID: 45
Enter Basic Salary: 50500
===EMPLOYEE DETAILS===
NAME: waleeja
EMPLOYEE ID: 35
BASIC SALARY: 40000
BONUS: 4000
FINAL SALARY: 44000
===EMPLOYEE DETAILS===
NAME: areej
EMPLOYEE ID: 45
BASIC SALARY: 50500
BONUS: 5050
FINAL SALARY: 55550
```